

TUKTOYAKTUK, NT, Canada

WMO: 719850

Lat: 69.436N

Lon: 133.023W

Elev: 6

StdP: 101.26

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.6	-37.1	-42.8	0.1	-38.2	-41.2	0.1	-36.9	18.4	-18.2	14.5	-15.8	1.7	90	0.973

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	22.6	15.6	20.3	14.5	17.8	13.5	16.4	21.1	15.1	19.3	13.8	17.4	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.1	18.5	13.0	9.3	17.0	11.8	8.6	15.8	46.2	20.9	42.3	19.4	38.8	17.4	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.3	9.1	DB	-40.7	26.5	2.0	2.1	-42.1	28.0	-43.2	29.3	-44.3	30.5	-45.7	32.0	(4)
(5)			WB	-40.8	18.4	1.9	1.5	-42.2	19.5	-43.3	20.4	-44.4	21.3	-45.8	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.8	-25.4	-25.4	-24.1	-14.6	-2.6	7.0	11.5	9.3	3.8	-5.9	-16.7	-23.2	(6)
(7)		DBStd	14.95	7.06	6.98	6.59	6.70	6.12	4.74	4.61	4.20	3.82	5.01	5.84	6.45	(7)
(8)		HDD10.0	7011	1098	991	1056	738	393	117	38	66	191	493	801	1029	(8)
(9)		HDD18.3	9899	1357	1224	1314	988	650	341	216	282	437	752	1051	1288	(9)
(10)		CDD10.0	158	0	0	0	0	1	27	84	43	3	0	0	0	(10)
(11)		CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0	(11)
(12)		CDH23.3	38	0	0	0	0	0	6	27	6	0	0	0	0	(12)
(13)		CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	4.4	4.3	4.4	4.0	4.5	4.6	4.6	4.7	4.7	4.4	4.2	4.0	(14)	
(15)	Precipitation	PrecAvg	136	5	5	3	6	12	22	31	17	16	7	7	(15)	
(16)		PrecMax	210	17	17	14	19	32	46	48	87	39	36	23	18	(16)
(17)		PrecMin	68	0	0	0	0	0	0	7	9	1	2	0	1	(17)
(18)		PrecStd	34	5	4	4	5	7	11	12	17	10	10	5	6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.0	-4.8	-4.1	3.2	16.4	23.7	25.9	23.6	17.2	5.3	-0.6	-5.5	(19)
(20)			MCWB	-3.3	-5.3	-4.8	1.6	9.8	14.8	16.4	17.1	11.9	4.1	-1.2	-5.9	(20)
(21)		2%	DB	-7.6	-9.0	-8.3	0.9	11.4	20.8	23.7	20.1	13.4	2.4	-4.1	-9.1	(21)
(22)			MCWB	-8.0	-9.5	-8.9	-0.3	6.9	13.6	16.1	15.1	9.7	1.5	-4.6	-9.4	(22)
(23)		5%	DB	-12.2	-12.0	-11.6	-1.5	7.7	18.0	21.5	17.6	10.8	1.1	-6.3	-11.6	(23)
(24)			MCWB	-12.5	-12.3	-12.0	-2.4	4.8	12.4	15.4	13.9	8.6	0.4	-6.8	-11.9	(24)
(25)		10%	DB	-15.8	-14.8	-14.5	-4.3	4.8	15.1	19.1	15.5	8.9	0.0	-8.6	-13.9	(25)
(26)			MCWB	-16.2	-15.1	-15.0	-4.9	2.8	10.8	14.6	12.7	7.4	-0.6	-9.1	-14.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.4	-5.2	-4.8	2.0	10.1	16.2	18.5	17.7	12.2	4.0	-1.3	-5.7	(27)
(28)			MCDWB	-3.1	-4.7	-4.2	3.1	15.9	22.0	23.8	22.8	16.5	5.1	-0.9	-5.5	(28)
(29)		2%	WB	-8.0	-9.4	-9.0	-0.2	7.2	14.1	17.0	15.6	10.2	1.7	-4.5	-9.4	(29)
(30)			MCDWB	-7.7	-9.1	-8.5	0.8	11.4	19.6	21.7	19.1	12.6	2.2	-4.2	-9.1	(30)
(31)		5%	WB	-12.5	-12.3	-12.0	-2.4	4.8	12.6	15.8	14.2	8.8	0.5	-6.8	-11.8	(31)
(32)			MCDWB	-12.2	-12.0	-11.6	-1.6	7.7	17.8	20.4	17.0	10.6	1.0	-6.4	-11.5	(32)
(33)		10%	WB	-16.2	-15.1	-15.0	-5.0	2.9	10.8	14.6	12.9	7.4	-0.6	-9.0	-14.2	(33)
(34)			MCDWB	-15.9	-14.9	-14.5	-4.3	4.6	15.1	18.7	15.1	8.9	-0.1	-8.6	-14.0	(34)
(35)	Mean Daily Temperature Range		MDBR	6.7	7.0	7.8	8.6	7.0	8.7	7.7	6.1	4.8	4.6	6.3	6.4	(35)
(36)		5% DB	MCDBR	10.3	9.6	9.2	8.9	11.9	14.1	11.8	9.7	8.4	3.9	7.0	9.0	(36)
(37)			MCWBR	10.2	9.5	9.0	8.2	7.9	8.2	5.8	5.5	5.4	3.5	7.0	9.0	(37)
(38)		5% WB	MCDWB	10.4	9.5	9.2	8.8	11.5	13.3	10.7	9.0	7.8	3.9	7.0	8.9	(38)
(39)			MCWBR	10.3	9.4	9.0	8.1	7.8	8.2	6.0	5.5	5.3	3.5	7.0	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.157	0.208	0.248	0.304	0.342	0.336	0.352	0.336	0.296	0.227	0.157	N/A	(40)	
(41)		taud	1.900	2.070	2.200	2.125	2.192	2.400	2.455	2.519	2.551	2.192	1.911	N/A	(41)	
(42)		Ebn at Noon	46	565	784	825	834	859	830	805	748	528	33	0	(42)	
(43)		Edh at Noon	4	40	72	108	115	97	90	75	55	34	3	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.02	0.41	2.04	4.32	6.10	6.42	5.13	3.22	1.67	0.52	0.06	0.00	(44)	
(45)		RadStd	0.00	0.04	0.09	0.20	0.34	0.48	0.43	0.31	0.15	0.05	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.99	+1.41	+1.75	N/A	N/A	N/A	-344	-358	N/A	N/A	(46)
(47) Regional (8 neighbors)		+1.04	+1.81	+2.05	N/A	N/A	N/A	-427	-422	N/A	N/A	(47)

Nomenclature: See separate page